LAYERS OF EARTH'S ATMOSPHERE

Mid-latitude values · heights not to scale (each layer stretched to fit)

HEIGHT
miles

EXOSPHERE (above): fades into space

373 mi — THERMOPAUSE

254 mi

160 mi

THERMOSPHERE

Temperature rises with height (solar UV and x-rays)

100 mi

Aurorae (62–250 mi)
green and red (highest): oxygen · blue, violet, and pink: nitrogen

Sky wave: shortwave radio
bounces off the ionosphere

International Space Station ~254 mi

Anacoustic zone: too thin for sound (99 mi)

62 mi — Kármán line, edge of space (62 mi)

Airglow (53–62 mi)
faint green glow from oxygen, seen even on moonless nights

53 mi — MESOPAUSE

Laser guide star
lights up sodium atoms, 56 mi

Noctilucent clouds (50–53 mi)

UVC UVB UVA

IONOSPHERE 37 → 373 mi

MESOSPHERE

Temperature falls with height

Red sprite
triggered by strong lightning below

40 mi

Meteoroids burn up as meteors

31 mi — STRATOPAUSE

STRATOSPHERE

Temperature rises with height (ozone absorbs UV)

25 mi

UVC: all absorbed by oxygen and ozone

20 mi

Blue jet

15 mi

SR-71 Blackbird 85,000 ft (16 mi)

Ozone layer (peak ~15.5 mi)

10 mi

Concorde 60,000 ft (11.4 mi)

Nacreous clouds
polar winter, 9–16 mi

UVB: mostly absorbed by ozone

6.8 mi — TROPOPAUSE

Thunderstorm anvil flattens at the tropopause

Jet stream → west to east, 80–140 mph

Airliner ~35,000 ft (6.6 mi)

Mt. Everest 29,032 ft (5.5 mi)

TROPOSPHERE

Temperature falls with height

4 mi

2 mi

0 mi

UVA reaches the ground

Observatory

Transmitter

Receiver

Weather: clouds, rain, lightning

0 ft = sea level

	TEMP °F	PRESSURE mb
Thermopause	~+2,000	~10^{-7}
	930 to 3,630 °F, depending on the Sun	
Space Station	~+1,650	~10^{-6}
Mesopause	−148	0.004
	−215 °F above the summer poles	
Stratopause	+28	1
Ozone peak	−61	25
Tropopause	−69	226
	about ¼ of sea-level pressure	
18,000 ft	−5	500
	half of all the air is below here	
Sea level	+59	1,013

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